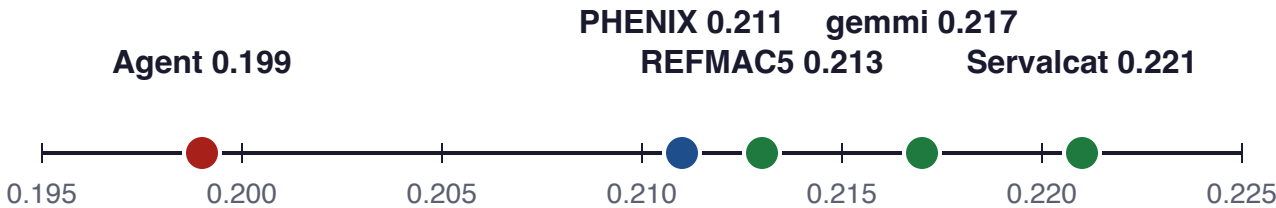


T03 — Reciprocal-space R-factors · 4 tools converge (Servalcat sigmaa added 2026-05-05)

Agent’s 0.199 R-free is 0.012 below the lower bound · 3 of 4 oracles fail R-free gap < 0.05

Tool	R-work	R-free	Gap
phenix.model_vs_data	0.156	0.211	0.055
gemmi sfcalc + custom	0.164	0.217	0.053
REFMAC5 (NCYC=0)	0.169	0.213	0.044
Servalcat sigmaa (NEW)	0.171	0.221	0.050
Agent (reported)	0.149	0.199	0.050

R-free across oracles



R-free under-statement is reproducible

Four independent code paths converge on R-free 0.211–0.221.
Likely cause: **phenix.refine in-run scaling vs catalog-oracle scaling**.
Mitigation: report from phenix.model_vs_data, not from refine’s log.
ASSUM_phenix_refine_in_run_r_factors_use_internal_scaling — known_violation hardened.

R-free gap criterion is scaling-dependent

PHENIX gap 0.055 (✗) · gemmi 0.053 (✗) · Servalcat 0.050 (✗) · REFMAC5 0.044 (✓)
Catalog oracle of record = phenix.model_vs_data → fails < 0.05 (3 of 4).
Reportable as a **scaling-sensitive failure**, not a raw R-free defect.